Unofficial English translation, made for demonstration purposes. Layout simplified; this is not the original document. Original: tijhuisingenieurs.nl/files/brochure-lunenburgerwaard-juli-2020.pdf

NEDERRIJN FLOODPLAINS

PROJECT PARTNERS

NATURE DEVELOPMENT LUNENBURGERWAARD

Floodplains of the Nederrijn: Nature Development in the Lunenburgerwaard

The Lunenburgerwaard already supports a great deal of wildlife, including special animals such as the beaver, the kingfisher and the badger, and special plants such as the melancholy thistle, the common restharrow and the black medick. With a few adjustments to the existing nature, there will be more opportunities to strengthen the area and make it even more attractive to plant and animal species (biodiversity).

Desired nature and landscaping measures in the Lunenburgerwaard

To achieve the desired nature, approximately 95 hectares of existing nature will be improved in 2021. Another important measure for further improving nature is its maintenance, that is, the application of management practices such as mowing. This takes place on land belonging to Utrechts Landschap and Rijkswaterstaat.

Corncrake

The Lunenburgerwaard is the floodplain area to the east of Wijk bij Duurstede. The province of Utrecht and Rijkswaterstaat see opportunities to strengthen nature while at the same time meeting the mandatory nature objectives of the European Natura 2000 network, the Netherlands Nature Network (Natuur Netwerk Nederland) and the Water Framework Directive, in order to improve water quality. With these obligations, we aim to create a more connected nature network along the floodplains.

Rijntakken Uiterwaarden Nederrijn

The Lunenburgerwaard forms part of the Natura 2000 site Rijntakken Uiterwaarden Nederrijn. This is the area between Heteren and Wijk bij Duurstede. During periods of high river discharge, the Nederrijn has to take on 1/6 of the discharge of the Rhine. During periods of low river discharge, the water is kept at level by the weir at Amerongen.

The floodplains vary in width and elevation and consist mainly of grassland interspersed with a few arable fields, hawthorn hedges, pollard willows, copses and small marshy areas.

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Water Framework Directive

Rijkswaterstaat is taking measures to improve ecological water quality. These measures stem from the European Water Framework Directive (WFD). The Directive stipulates that waters must provide a good habitat for the plants and animals that belong there. We want surface waters to function well ecologically again. This sustainable approach offers opportunities for the Netherlands, its citizens and our business community. There are still significant gains to be made by restoring habitats, giving fish free passage and keeping the water clean and healthy. Water flows from top to bottom and does not respect borders. For this reason, Rijkswaterstaat works closely with the other Dutch water managers and neighbouring countries to achieve this goal.

What Natura 2000 means

Biodiversity (species richness) in Europe has been declining for years. Sustainable protection of plants and animals is urgently needed. Plants and animals pay little heed to national borders, which is why it is important to approach nature conservation at a European level. In this way, we prevent nature in Europe and in the Netherlands from becoming increasingly uniform. To this end, the Birds Directive was drawn up in 1979 and the Habitats Directive in 1992. These directives have two components: species protection and area protection. All EU member states designate protected areas for specific (habitats of) (bird) species. The protected areas designated under both directives form the Natura 2000 network. The Dutch contribution to this European network of protected nature areas consists of more than 160 areas. These areas are located both on land and at sea. Most of the areas have now been definitively designated.

Natura 2000 areas within the province of Utrecht and Lunenburgerwaard highlighted

Nature Network Netherlands

The Nature Network Netherlands (Natuur Netwerk Nederland, NNN) was established after nature had been in decline in the Netherlands for many years. Many nature areas had become fragmented, which meant that many plants and animals were doing poorly. Animals in particular had insufficient habitat in which to reproduce. Since 2013, Nature Network Netherlands has been the name of the Netherlands' ecological main structure: a coherent network of existing and future nature areas in the Netherlands. It forms an important part of nature policy. The aim is at least to stabilise biodiversity in the Netherlands, and thus to counter further decline.

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This page describes the measures that are being applied to achieve the desired nature. The measures have no effect on the other functions in the area.

blikonderwater, underwater photography

MARSH AND FRESHWATER LAKE

LOCATION: PLASDRAS GRAVENBOL

Desired nature: A lake with a broad marsh will be created at Gravenbol. This is a shallow channel with low-growing plants (sedge vegetation) which is connected to the old river arm (Dode Arm) at high water. Part of the marsh will be permanently under water. The lake and the marsh will contribute to the development of various species of fish, such as tench, weatherfish and bitterling, as well as dragonflies and many bird species, ranging from waders to spotted crake, common snipe, ducks and geese.

The lake also strengthens the quality of the current natural values, because the area is already an important breeding area for several species under Natura 2000.

Intervention: Excavation of a lake and a high-water inflow opening, in the form of a sill, at the old river arm.

Management: The plants around the marsh are sensitive to grazing. The plants are therefore fenced off in spring and summer, with the possibility of aftermath grazing in autumn. An inspection takes place annually to determine whether mowing is required. The management of the surrounding natural grasslands remains unchanged.

NATURE-FRIENDLY BANKS

LOCATION: SOUTH BANK OF THE OLD RIVER ARM

Desired nature: On the south bank of the old river arm, the stone groyne fields have been eroded by wave action in a number of places, resulting in sandy, shallow, gently sloping banks.

It is desirable to continue this at the locations where rock armour is still present, in order to create shallow water. The stones provide shelter for juvenile fish, and algae and macrofauna can live on them. Waders forage on the sandy banks, and the green clubtail (a dragonfly) finds a place to lay its eggs. The water of the old river arm is a foraging area for waterfowl such as great crested grebe, cormorant, various ducks and geese.

Intervention: The stones are pushed into the water, so that the bank is subject to less wave action. The banks will develop further through natural erosion.

Management: Management is required if too much erosion occurs. This is checked periodically.

Natural erosion of the bank

RIVER WOOD

LOCATION: EAST OF THE OLD RIVER ARM

Desired nature: Dead wood in the water, lying largely below the water surface, forms a nutrient-rich substrate for all kinds of aquatic organisms (macrofauna) and provides shelter for growing juvenile

fish. Fish-eating birds (great crested grebe and cormorant) in turn benefit from this, and altogether it contributes to the biodiversity of the area.

Intervention: Dead trees (20 in total) are placed in the least navigated part of the river arm. This causes little nuisance to recreation such as pleasure boating. The dead wood is placed near the old groyne fields, creating a wind-free area. This area is created between the river wood and the bank line and can serve as habitat for fish and promote the development of bank and water plants.

Management: Anchoring is required to ensure that the river wood stays in place. Periodic checks are carried out for safety and to inspect the anchoring of the river wood.

SANDENBURGERWAARD

Desired nature: Once the excavation in the Sandenburgerwaard is complete, an area of pools, marshland and grassland will remain. At high water, the pools can occasionally be connected to the old river branch. Because the connection is limited, there is less dynamism in the water, which creates more opportunities for clear water with rich aquatic vegetation to develop. The drier parts will develop into false oat-grass hay meadows.

Intervention: To create this periodic connection, a high-water inflow opening will be made to the old river branch in the form of a sill. On the basis of the annual water levels, a calculation has been made of how high this sill must be built to ensure sufficient exchange of water. The connection with the river branch occurs less often than at the Gravenbol inundated area. The Sandenburgerwaard is deeper than the marshland at Gravenbol.

Management: It is important that the sill retains the correct shape for the inflow of water. Periodic inspections will be carried out.

MOIST HAY MEADOW AND FALSE OAT-GRASS HAY MEADOW

LOCATION: FORMER AGRICULTURAL PLOT

Desired nature: In the agricultural area to the west of the Sandenburgerwaard, south of the dyke, grasslands will be developed with partly wet and partly dry substrate. This area was not excavated by the brickworks, unlike the surrounding area where soil was dug out (and contaminated soil was dumped back) for the manufacture of bricks. This makes the area promising for developing flower-rich grasslands (promoting biodiversity) with species such as yellow rattle, common bird's-foot trefoil, greater bird's-foot trefoil, lady's bedstraw, marsh marigold and various sedge species.

Intervention: By removing the nutrient-rich top layer, the conditions for lean and moist hay meadow are created. The desired species will be introduced using green hay from other locations.

Management: Management consists of mowing and removing the grass, with no fertilisation.

DRY GRASSLAND ON POOR SOIL

LOCATION: NORTH BANK OF RIVER BRANCH

Desired nature: Dry, nutrient-poor grassland will be developed on the higher parts of the area (channel ridges, natural levee, summer embankment). This is a characteristic river-related grassland type that is important for various insects, such as bees.

A number of characteristic species of nutrient-poor grassland are already present in the area, such as spiny restharrow and smooth rupturewort. To make it even more attractive to insects, the transition from nutrient-poor grassland to shrubland (scrub) will be improved and strengthened.

Intervention: Planting of various shrub species along the existing woodland edge, to the east of the brickworks. The shrubland will consist of shrubs 1 to 5 metres high.

Management: The dry nutrient-poor grassland will be grazed and not fertilised. In the shrubland, young trees that grow spontaneously will be cut down once every 10 years.

Desired situation for the area

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HERB- AND FAUNA-RICH GRASSLAND

LOCATION: VARIOUS GRASSLANDS

Desired nature: To convert various grasslands into grassland with more flowers, such as buttercup, common sorrel and oxeye daisy. Many insects and birds benefit from the flowers.

Intervention: Only management measures will be applied.

Management: Management consists of mowing and removing the grass. No fertilisation.

NATURE FIELD, SCRUB ZONE AND TALL HERB FIELD

LOCATION: EAST OF THE MARINA

Desired nature: To develop more herb-rich fields. At the transitions to the forest, this will be combined with an area of shrub species and flowering tall herbs to promote plant diversity. Birds, butterflies, insects, bats and the badger benefit from this.

Intervention: By planting various shrub species along the existing forest edge, a scrub edge will be created. A scrub edge consists of shrubs 1 to 5 metres high.

Management: The nature field will be sown in early spring and possibly harvested in autumn. The tall herbs will be mown in autumn, annually or every two years, depending on development/encroachment of woodland. In the scrub zone, shrubs that develop into trees will be removed occasionally.

Captions: Nature field; Tall herbs and scrub

RIVERINE FOREST

LOCATION: EXISTING FOREST AND NEW FOREST TO BE PLANTED

Desired nature: The existing and highly valuable softwood floodplain forest around the brickworks will be retained and strengthened through expansion. This is the habitat of special plant and animal species, such as the golden oriole, the turtle dove, the nightingale and the beaver. Floodplain forest is forest that grows in the floodplains and can vary from moist to wet softwood floodplain forest, and hardwood floodplain forest on the drier parts. It is unique in the Netherlands to retain forest in the floodplain and to be permitted to expand it, in view of the river's flow during high water.

Intervention: Planting of a plot with native tree species, such as maple, poplar and lime.

Management: For the first ten years, the forest will be managed to allow it to develop well. When trees die off, new trees are sometimes planted, or trees are removed to give space to other trees or shrubs.

Caption: Current situation of the riverine forest

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CONTROL OF GIANT HOGWEED IN FOREST EDGE ZONES

LOCATION: SUMMER DYKE AND FOREST EDGES

Desired nature: A highly dominant exotic species, giant hogweed, occurs in the forest edge areas. This exotic species hinders the development of the forest edge areas. By controlling the giant hogweed, native species of shrubs and tall herbs will have a better chance and a more natural transition will emerge.

Intervention: To control the giant hogweed, the hogweed must be dug out and mown over a period of 2 years.

DIVERTING A DITCH

Desired nature: To limit desiccation of the woodland areas and to guarantee drainage of the remaining agricultural areas, a ditch will be diverted. Diverting the ditch creates new development opportunities for the existing woodland.

Intervention: To the north-west of the brickworks, the ditch will be diverted in a northerly direction. This new ditch will take over the function of the section that falls out of use.

ACCESSIBLE NATURE – WALKING ROUTE

The plan includes a walking route. This makes the area partially accessible to walkers who wish to enjoy nature. The walking route starts in Wijk bij Duurstede and takes walkers past the various types of nature with which the Lunenburgerwaard is richly endowed: the nutrient-poor grasslands, the woodlands and the waterside areas. The walking route is an attempt to improve the opportunities for walking in the Lunenburgerwaard without causing damage to vulnerable nature.

An assessment has been made of whether the layout of the walking route affects various protected species (including birds) that occur, or may occur, in the area. It is important that parts of the nature reserve are left undisturbed. To this end, a quiet zone has been created in the eastern part of the plan area; this area is therefore not accessible to recreational visitors.

OTHER MEASURES

The plot where the ditch will be dug

FINAL LAYOUT OF NATURE DEVELOPMENT AT LUNENBURGERWAARD, AFTER 2022

This shows the final layout for the nature development in the floodplain. All the measures mentioned previously are included. The design of the nature in the dyke zone is not yet known and has therefore not been included.

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EINDBEELD NATUURONTWIKKELING LUNENBURGERWAARD, NA 2022# FLOODPLAINS NEDERRIJN NATURE DEVELOPMENT LUNENBURGERWAARD 9

FINAL STATE OF NATURE DEVELOPMENT AT LUNENBURGERWAARD, AFTER 2022

OTHER DEVELOPMENTS IN THE FLOODPLAIN

A great deal is currently happening in the floodplain, such as the reinforcement of the Lekdijk and the sand extraction in Sandenburgerwaard. The Province of Utrecht is in close contact with all parties about the works and the interface with these projects, which are not part of this project. The map below shows all projects in the area, together with the name of the contractor or owner. For the desired development, the relocation of the Gravenbol day recreation area, no detailed plan is yet known.

View of the Lunenburgerwaard from Wijk bij Duurstede

Contact and questions

Do you have any questions about the nature development of the Lunenburgerwaard, or would you like more information? Send an email to avp@provincie-utrecht.nl or a letter to Provincie Utrecht, Postbus 80300, 3508 TH Utrecht, marked for the attention of Natuurontwikkeling Lunenburgerwaard.

Documents

More information about the nature development and the design is available on our website: www.provincie-utrecht.nl/nederrijn

Planning

The permitting process must be completed before the nature development can be carried out. This will take place in the autumn of 2020. The nature restoration in the floodplain will be carried out in mid-2021. Hoogheemraadschap De Stichtse Rijnlanden will include the nature restoration in the dyke zone in the Wijk - Amerongen Dyke Reinforcement project. The works in the dyke zone are scheduled for 2023.

PLANNING

Timeline: 2019 - 1st half of 2020 | 2020 | 1st half of 2021 | 2nd half of 2021

Phases: Exploration | Plan development | Realisation

Preparation of sketch design (SD)

Preparation of preliminary design (PD)

Communication on PD (including information meetings, one-to-one discussions and newsletter)

Exploratory studies (including soil, environment, nature)

Preparation of final design (FD)

Further studies (including nature, soil, environment, river engineering, archaeology)

Communication on PD (including information meeting, one-to-one discussions, newsletter)

Adoption of FD

Communication on FD (including one-to-one discussions, website, newsletter)

Permit applications

Public inspection of permits, draft decisions (6 weeks)

Preparation of specification

Public inspection of permits, final decisions (6 weeks)

Communication on permitting procedure (informing about the procedure)

Adoption of specification

Tendering for realisation

Works in progress (approximately 6 months)

Construction of the nature area

Nature management starts from 2022.

The following organisations are involved in this project: the Province of Utrecht, Rijkswaterstaat, Utrechts Landschap, the municipality of Wijk bij Duurstede and Hoogheemraadschap De Stichtse Rijnlanden.

More information about this project can be found on the website: www.provincie-utrecht.nl/nederrijn.

Questions about this project can be sent by email to the Province of Utrecht, Living Environment department, team AVP: avp@provincie-utrecht.nl

Information about HDSR's adjacent dyke reinforcement project can be found at: www.destichtserijnlanden.nl/sterkelekdijk

July 2020